Appendix

TABLE I
EXPLANATION OF THE SYMBOLS USED IN THE PAPER.

Used Symbols	Descriptions
CAS	Central Aggregation Server
EoCL	"Ensemble of Classifiers" with m classifiers either in series or in parallel
U and τ	Global and local iterations respectively
$\mathcal{E}$	Globally trained EoCL
G	Globally trained learning model
$C_i : i = 1 \cdots n$	Local edge clients
$\mathcal{E}_i : i = 1 \cdots n$	Local replica of global EoCL at C_is
$G_i : i = 1 \cdots n$	Local replica of G
$\mathcal{B}_i : i = 1 \cdots n$	Private data of active C_i
$\mathcal{B}_i^+ : i = 1 \cdots n$	Filtered (using using local EoCL) private data of active C_i
$Cl_j : j = 1 \cdots N$	Classes in the multi-modal training data
$Cl_j M_i : j = 1 \cdots N;\ i = 1 \cdots m$	i-th classifier model trained on the i-th partition of Cl_j
$\mathcal{T}$	Pristine training set
$\mathcal{T}^\circ$	Poisoned training set
$\delta\alpha_j$	Accuracy metric for class j or Cl_j
$E(C_i) = \{D_i^u, D_i^t\}$	Execution environment in active edge C_i
D_i^t and D_i^u	Trusted enclave and untrusted domain in C_i
H_i^{BIN} and H_i^{CFG}	Hash of executable binaries, and reference hash of approved configuration runtime respectively in C_i
$\mathcal{A}$	A set potential adversaries
$\nabla_{\theta_i} G_i$	Local gradient in C_i
$\mathcal{H}$	A set of potential active honest edges
IID	Independent and Identically Distributed, i.e., Data samples are independent and drawn from the same distribution
$\beta \leq 1$	Poison fraction, i.e., fraction of clients among the total number of active edges are corrupted

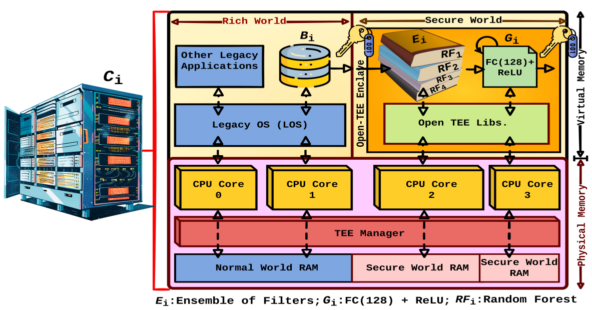

Fig. 1. Overview of the secure enclave in any active edge. Here, $\mathcal{E}_i$ contains four RFs with different parameter settings and the local G_i as FC(128)+ReLU.

TABLE II
PARAMETER SETTINGS DETAILS (FOR REPRODUCIBILITY) OF THE EOCLS USED IN THE EXPERIMENTS. NOTE: ALL PARAMETERS ARE SCIKIT-LEARN V1.6.1 DEFAULTS UNLESS BOLDED; BOLD = VALUES EXPLICITLY SET IN OUR RUNS.

Parameter	**Preprocess**	**SVM**	**RF**	**MLP**
with_mean	**True**	—	—	**True**
with_std	**True**	—	—	**True**
copy	True	—	—	True
C	—	**4.0**	—	—
kernel	—	rbf	—	—
gamma	—	scale	—	—
degree	—	3	—	—
coef0	—	0.0	—	—
shrinking	—	True	—	—
probability	—	False	—	—
tol	—	1e-3	—	1e-4
cache_size	—	200	—	—
class_weight	—	None	—	—
verbose	—	False	0	False
max_iter	—	**-1**	—	**30**
decision_function_shape	—	ovr	—	—
break_ties	—	False	—	—
random_state	—	**42**	**42**	**42**
n_estimators	—	—	**200**	—
criterion	—	—	gini	—
max_depth	—	—	None	—
min_samples_split	—	—	2	—
min_samples_leaf	—	—	1	—
min_weight_fraction_leaf	—	—	0.0	—
max_features	—	—	sqrt	—
max_leaf_nodes	—	—	None	—
min_impurity_decrease	—	—	0.0	—
bootstrap	—	—	True	—
oob_score	—	—	False	—
n_jobs	—	—	-1	—
warm_start	—	—	False	False
ccp_alpha	—	—	0.0	—
max_samples	—	—	None	—
hidden_layer_sizes	—	—	—	**(256, 128)**
activation	—	—	—	relu
solver	—	—	—	adam
alpha	—	—	—	1e-4
batch_size	—	—	—	**256**
learning_rate	—	—	—	constant
learning_rate_init	—	—	—	0.001
power_t	—	—	—	0.5
shuffle	—	—	—	True
early_stopping	—	—	—	**True**
validation_fraction	—	—	—	0.1
momentum	—	—	—	0.9
nesterovs_momentum	—	—	—	True
beta_1	—	—	—	0.9
beta_2	—	—	—	0.999
epsilon	—	—	—	1e-8
n_iter_no_change	—	—	—	3
max_fun	—	—	—	15000

A. Proof of Theorem 1

Let us partition the active edges $\{C_i\}_{i=1}^{M}$ into honest ($\mathcal{H}$, with $|\mathcal{H}| = (1-\beta)M$) and adversarial ($\mathcal{A} = \beta M$) subsets. Then, $\theta_{\text{GLOBAL}}^{t+1} = \big(1-\beta\big)\hat{\theta}_{\mathcal{H}}^{t+1} + \beta\hat{\theta}_{\mathcal{A}}^{t+1}$, where $\hat{\theta}_{\mathcal{H}}^{t+1} = \frac{1}{(1-\beta)M}\sum_{i\in\mathcal{H}} w_i \mathcal{E}_i\big(\theta_i^{t+1}\big)$ and analogously for $\hat{\theta}_{\mathcal{A}}^{t+1}$. The clean attack-free aggregation is $\theta_{\text{CLEAN}}^{t+1} = \hat{\theta}_{\mathcal{H}}^{t+1}$.

Subtracting we get

$$\left\|\theta_{\text{GLOBAL}}^{t+1} - \theta_{\text{CLEAN}}^{t+1}\right\|_2 = \beta \left\|\hat{\theta}_{\mathcal{A}}^{t+1} - \hat{\theta}_{\mathcal{H}}^{t+1}\right\|. \tag{1}$$

For any adversarial client $i \in \mathcal{A}$,

$$\begin{aligned}\left\|\mathcal{E}_i\left(\theta_i^{t+1}\right) - \hat{\theta}_i^{t+1}\right\|_2 &\leq \left\|\theta_i^{t+1} - \hat{\theta}_i^{t+1}\right\|_2 + \left\|\mathcal{E}_i\left(\theta_i^{t+1}\right) - \theta_i^{t+1}\right\|_2 \\ &\leq \Delta + \delta_{\alpha_i} \leq 2\Delta\end{aligned} \tag{2}$$

Therefore, by Jensen's inequality for convex norms,

$$\left\|\hat{\theta}_{\mathcal{A}}^{t+1} - \hat{\theta}_{\mathcal{H}}^{t+1}\right\|_2 \leq 2\Delta \tag{3}$$

Substituting Eq. 3 in to Eq. 1 we get the following:

$$\left\|\theta_{\text{GLOBAL}}^{t+1} - \theta_{\text{CLEAN}}^{t+1}\right\|_2 \leq 2\beta\Delta. \tag{4}$$

For weighted averaging with normalization $\frac{1}{1-\beta}$ to preserve unbiasedness, the tight constant becomes $\frac{2\beta}{1-\beta}$, giving the final bound. If the global objective $\mathcal{U}(\theta)$ is L-smooth,

$$\begin{aligned}\left|\mathcal{U}\left(\theta_{\text{GLOBAL}}^{t+1}\right) - \mathcal{U}\left(\theta_{\text{CLEAN}}^{t+1}\right)\right| &\leq I \left\|\theta_{\text{GLOBAL}}^{t+1} - \theta_{\text{CLEAN}}^{t+1}\right\|_2 \\ &\leq \frac{2I\beta}{1-\beta}\Delta,\end{aligned} \tag{5}$$

establishing bounded loss degradation and stability of subsequent updates.

1) Interpretation & Insight: (*1*) *Admissibility as trust region*: The filter $\mathcal{E}_i$ projects each update into a convex "safe zone" of radius Δ around the honest manifold; no local update can pull the global model outside this region; (*2*) *Linear robustness law*: Global deviation grows linearly with attack fraction β and local divergence Δ, giving a measurable security budget: halving Δ or β halves the deviation; (*3*) *Tightness*: The constant $\frac{1}{1-\beta}$ is tight for affine aggregation under worst-case poisoning; stronger filters (*e.g.*, trimmed-mean EoCL) yield smaller empirical constants; and (*4*) *System relevance*: The bound translates to a Zero-Trust contract: if every client attestation ensures Δ-bounded admissibility, then the entire FL process preserves integrity within an analytically provable envelope.

B. Proof of Theorem 2

The execution environment $E(C_i)$ in an active edge C_i is defined as $E(C_i) = \{D_i^u, D_i^t\}$, where D_i^u and D_i^t, respectively, the untrusted (fully observable and controllable by adversary $\mathcal{A}$) and trusted environments in an active edge C_i. The local secret tuple $\mathcal{S}_i = \{\theta_i, \nabla_{\theta_i} G_i, \mathcal{E}_i\}$ resides entirely within D_i^t. The trusted domain runs a measured boot chain: $\text{CAS} \longrightarrow_{\text{ATTESTATION}} D_i^t$, producing a signed quote $\sigma_i = \text{Sign}_{\text{CAS}}\left(H(D_i^t)\right)$. Only if $H(D_i^t)$ matches the authorized policy manifest $\mathcal{P}_i$ does the enclave receive its session keys and policy tokens.

1) Confidentiality Argument: Let K_i be the symmetric encryption key provisioned by CAS after attestation. All enclave state variables—θ_i, $\nabla_{\theta_i} G_i$ and classifier ensemble $\mathcal{E}_i$ are stored and transmitted as $C_i = \text{Enc}_{K_i}(\mathcal{S}_i \| r)$, where r is nonce-based randomness. Even if $\mathcal{A}$ intercepts or copies C_i through DMA, I/O, or host memory inspection, it cannot recover $\mathcal{S}_i$ without K_i, since $\mathbb{P}\left[\text{Dec}_{K_i'}(c_i) = \mathcal{S}_i\right] \le 2^{-\lambda}$, with λ denoting the cryptographic security parameter (*e.g.*, 128-bit AES). Thus, data confidentiality holds up to the TEE's residual side-channel leakage ℓ_{TEE}.

2) Integrity Argument: By the measured-boot chain, any modification of the enclave binaries or configuration changes $H(D_i^t)$. Because the CAS signs only verified measurements, the modified enclave fails attestation, and consequently:

$$K_i \notin D_i^t; \text{ hence, } \mathcal{S}_i \text{ is never materialized} \tag{6}$$

Formally, for any modification operator Δ applied by $\mathcal{A}$,

$$\mathbb{P}\left[\text{Attest}\left(\mathcal{P}_i \oplus \Delta\right) = \text{Valid}\right] = 0. \tag{7}$$

Therefore, $\mathcal{A}$ cannot inject or alter θ_i, $\nabla_{\theta_i} G_i$, and / or $\mathcal{E}_i$ within an attested enclave instance.

3) Channel and Leakage Bound: Let the total leakage channels available to $\mathcal{A}$ be:

$$\mathcal{L} = \mathcal{L}_{\text{CACHE}} \bigcup \mathcal{L}_{\text{TIMING}} \bigcup \mathcal{L}_{\text{SPECULATIVE}} \tag{8}$$

For modern enclaves (*e.g.*, Intel SGX, AMD SEV, ARM CCA), mitigations bound observable information leakage by a negligible function $\ell_{\text{TEE}} = f(t, \nu)$, where t is the timing resolution, and ν is the noise entropy. Thus, the adversary's advantage satisfies:

$$\text{Adv}_{\mathcal{A}}^{\text{TEE}} = \left|\mathbb{P}\left[\mathcal{A} \to \mathcal{S}_i\right] - \frac{1}{2}\right| \le \ell_{\text{TEE}}. \tag{9}$$

In other words, $\mathcal{A}$'s posterior knowledge of $\mathcal{S}_i$ is statistically indistinguishable from random guessing.

4) Composition with Secure Channel: Communications between enclave D_i^t and server or other enclaves are authenticated via mutually attested TLS (mTLS) using session key K_i^{SESS} bound to the enclave identity. Hence, even if $\mathcal{A}$ mounts a man-in-the-middle attack within D_i^u, the channel transcript leakage remains bounded by:

$$\mathbb{P}\left[\text{Forge}_{\mathcal{A}}\left(K_i^{\text{SESS}}\right)\right] \le 2^{-\lambda}. \tag{10}$$

Therefore, no feasible strategy exists for $\mathcal{A}$ to extract or alter $\mathcal{S}_i$ during transmission or at rest.

Next, combining the confidentiality, integrity, and leakage bounds, we have:

$$\mathbb{P}\left[\mathcal{A} \to \mathcal{S}_i\right] \le \ell_{\text{TEE}} + 2^{-\lambda} \approx \ell_{\text{TEE}}. \tag{11}$$

Hence, under the stated assumptions of a trusted CAS root and a correctly implemented TEE, the adversary's success probability in obtaining or modifying any element of $\mathcal{S}_i$ is negligible. Hence, proved.

TABLE III
SFL FRAMEWORK GENERALIZES OR OUTPERFORMS FLCERT, CROWDGUARD, AND PPFL IN ROBUSTNESS AND TRUST ENFORCEMENT.

Dimension	SFL [Ours]	FLCERT	CROWDGUARD	PPFL
Primary Defense Locus	Client + Server (end-to-end)	Server (certified robust aggregation)	Server (federated backdoor detection)	Client (TEE isolation) + Server
Adversary Model	Bounded malicious fraction β adaptive / stealthy poisoning, model theft, MITM	Byzantine/poisoning; mostly update-level	Backdoors / poisoning (coordinated)	Host compromise vs. TEE; honest -but-curious
Client-side Input Filtering	Yes; ensemble AEOCL (pre-aggregation)	No	No (post-hoc server detection)	No
Enclave-Protected Training	Yes; model + filter in TEE (measured boot, attestation)	No	No	Yes; model in TEE
Confidentiality / Integrity / Authenticity	All three (attested channels, signed deltas, integrity chain)	Integrity (aggregation certificate); weak on confidentiality / authenticity	Integrity focus (detector); limited confidentiality / authenticity	Confidentiality+ integrity via TEE; limited authenticity
Forensic Visibility (Auditability)	High; provenance logs, trust scores, rollback	Moderate; certified bounds, limited telemetry	Moderate; detector outputs	Low–to–Mod; TEE opacity, limited server insight
Robustness Mechanism	Layered: pre-filter + ZT gate + robust aggregation	Certified robust aggregation	Backdoor detection / triage	Isolation (TEE) + standard FL
Defense Against Adaptive/Covert Attacks	Strong; refreshable ensembles + behavior models	Moderate; bounded by cert assumptions	Moderate; detector evasion possible	Limited; TEE $\neq$ data poisoning defense
Privacy Options (DP / SecAgg)	DP-SGD optional; keeps forensics (no blind SECAGG)	Compatible; may reduce detectability	Compatible; may reduce visibility	Compatible; enclave may reduce DP need
Overhead on Edge	Low-to-Moderate; (lightweight ensembles; enclave-optimized)	Low; (server-heavy)	Low-to-Moderate; (server-side compute)	Moderate; (enclave context + I/O)
Scalability (Heterogeneous IoT)	High; designed for resource-variance	High; server-centric	High; server-centric	Moderate-to-High; TEE / device constraints

TABLE IV

ACCURACY OF DIFFERENT FILTER ENSEMBLES (USING A VARYING NUMBER OF CLASSIFIERS OR m) AGAINST VARIOUS CORRUPTED TRAINING DATA (INCLUDING MNIST, GTSRB, AND LSUN-BEDROOM) CONTAINING 5%-20% POISONED SAMPLES (PREPARED USING *mean-shift perturbations*) PER CLASS, I.E., $\rho \in (0.05, 0.2)$. THIS COMPUTATION IS AVERAGED OVER 80 ACTIVE C_iS AFTER 5000 GLOBAL ITERATIONS.

MNIST							GTSRB						
ρ	⟵ No. of RFs ⟶			⟵ No. of SVMs ⟶			ρ	⟵ No. of RFs ⟶			⟵ No. of SVMs ⟶		
↓	$m=1$	$m=3$	$m=5$	$m=1$	$m=3$	$m=5$	↓	$m=1$	$m=3$	$m=5$	$m=1$	$m=3$	$m=5$
0	0.000	0.000	0.000	0.000	0.000	0.000	0	0.000	0.000	0.000	0.000	0.000	0.000
	96.091	96.011	93.332	95.712	93.011	92.125		97.312	95.112	94.019	96.089	95.219	94.001
0.05	-3.717	-1.121	-0.239	-3.921	-1.772	-0.301	0.05	-3.092	-1.201	-0.409	-3.091	-1.001	-0.891
	92.374	**94.890**	93.093	91.791	91.239	91.824		**94.229**	93.911	93.610	92.998	94.219	93.110
0.10	-2.912	-1.315	-0.911	-3.089	-1.904	-0.921	0.10	-7.219	-5.331	-1.619	-7.998	-5.582	-1.658
	93.179	**94.696**	92.421	92.623	91.107	91.204		90.102	89.781	**92.400**	88.091	89.637	92.343
0.15	-10.711	-6.011	-3.089	-11.981	-7.790	-5.101	0.15	-10.18	-9.39	-5.87	-10.91	-9.22	−4.91
	85.38	90.000	**90.243**	83.731	85.221	87.024		87.141	85.722	88.149	85.179	85.999	**89.0911**
0.20	-35.091	-15.011	-4.091	-34.981	-16.011	-6.101	0.20	-38.31	-14.31	-9.56	-35.11	-18.89	-8.81
	61.000	81.000	**89.241**	60.731	77.000	86.024		59.011	80.802	84.459	60.979	76.32	**85.191**

LSUN-Bedroom													
ρ	⟵ No. of RFs⟶						ρ	⟵ No. of SVMs ⟶					
↓	$m=1$	$m=3$	$m=5$	$m=9$	$m=11$	$m=21$	↓	$m=1$	$m=3$	$m=5$	$m=9$	$m=11$	$m=21$
0	0.000	0.000	0.000	0.000	0.000	0.000	0	0.000	0.000	0.000	0.000	0.000	0.000
	98.347	97.134	96.181	96.141	94.118	93.165		98.211	97.399	97.115	95.482	95.678	94.117
0.05	-3.471	-2.691	-2.582	-1.412	-1.881	-0.813	0.05	-3.891	-3.486	-2.856	-2.718	-1.912	-0.881
	94.876	95.443	**95.599**	94.729	92.237	92.352		**94.320**	93.913	94.259	92.710	93.766	93.236
0.10	-7.991	-6.682	-6.497	-4.678	-1.991	-1.826	0.10	-8.761	-8.210	-7.391	-5.861	-5.136	-5.021
	90.356	**92.452**	89.684	91.463	92.127	91.399		89.450	89.189	89.724	89.567	90.544	89.096
0.15	-14.48	-9.57	-7.87	-7.23	-4.01	-3.09	0.15	-18.367	-16.091	-12.231	-12.094	-11.119	-11.998
	83.867	87.564	88.311	88.911	**90.108**	90.075		79.844	81.308	**84.884**	83.334	84.559	82.119
0.20	-38.091	-19.183	-15.177	-9.079	-6.157	-6.153	0.20	-37.511	-26.998	-17.091	-17.331	-10.119	-9.335
	60.256	77.951	81.004	87.062	**87.961**	87.012		60.700	**70.401**	80.024	78.097	85.559	84.782

TABLE V

ACCURACY OF THE MODELS ON THE PRISTINE BASELINE DATASETS BOTH WITH IID AND NON-IID ASSUMPTIONS. BLUE AND RED HIGHLIGHTS ARE RESPECTIVELY THE VALUES CORRESPONDING TO THE BEST MODELS FOR IID AND NON-IID ASSUMPTIONS.

On Pristine MNIST: Accuracy 97.0% (Approx.)						
Models	τ	α	U	IID	1-C	5-C
	1	10	4591	321	212	102
Conv(128,6,	**5**	**100**	**4510**	**31**	144	150
	1	10	4315	179	223	70
Conv(128,5,	5	100	5430	473	145	67
	1	10	5430	175	112	69
Conv(64,5,5	5	100	4560	155	241	72
	1	10	4150	212	134	112
FC(300)+ReL	**5**	**100**	**4280**	200	**64**	**53**
	1	10	4200	193	122	113
FC(128)+ReL	5	100	3500	39	187	127

On Pristine CIFAR-10: Accuracy 98.0% (Approx.)						
Models	τ	α	U	IID	1-C	5-C
	1	10	5300	124	142	112
Conv(128,6,6)+ReLU	**5**	**100**	**5100**	**41**	190	165
	1	10	5120	109	161	113
Conv(128,5,5)+ReLU	**5**	**100**	**5000**	116	**72**	**38**
	1	10	5000	120	101	131
Conv(64,5,5)+ReLU	5	100	4100	45	103	112
	1	10	5050	111	142	121
FC(300)+ReLU	5	100	5280	131	109	145
	1	10	5200	170	169	180
FC(128)+ReLU	5	100	5500	117	104	182

On Pristine LSUN-Bedroom: Accuracy 98.0% (Approx.)						
Models	τ	α	U	IID	1-C	5-C
	1	10	6900	296	212	134
Conv(128,6,6)+ReLU	**5**	**100**	**3810**	**25**	100	121
	1	10	5715	379	221	159
Conv(128,5,5)+ReLU	5	100	7320	213	192	202
	1	10	5912	148	161	187
Conv(64,5,5)+ReLU	5	100	6123	173	170	152
	1	10	6150	217	321	302
FC(300)+ReLU	5	100	6280	311	213	219
	1	10	1200	252	191	129
FC(128)+ReLU	**5**	**100**	**3500**	87	**34**	**29**

On Pristine GTSRB: Accuracy 98.0% (Approx.)						
Models	τ	α	U	IID	1-C	5-C
	1	10	5141	110	103	132
Conv(128,6,	**5**	**100**	**5310**	**21**	**35**	**39**
	1	10	5315	119	112	110
Conv(128,5,	5	100	5130	151	172	192
	1	10	5150	141	109	123
Conv(64,5,5	5	100	5160	191	132	123
	1	10	5150	412	201	218
FC(300)+ReL	5	100	5280	143	138	182
	1	10	5200	158	206	209
FC(128)+ReL	5	100	5500	137	123	141

TABLE VI

ACCURACY OF BASELINE MODELS ON POISONED DATASETS UNDER IID AND NON-IID ASSUMPTIONS WITH 9 CORRUPTED CLIENTS. BLUE AND RED INDICATE THE BEST MODELS FOR IID AND NON-IID, RESPECTIVELY.

On Corrupted MNIST with $^1\epsilon = 0.05$; $^2\epsilon = 0.20$							
Models	τ	γ	U	IID^1	IID^2	1-C^1	5-C^2
	1	10	5100	110	101	117	119
Conv(	**5**	**100**	**5100**	**49**	**58**	115	109
	ReLU 1	10	5100	117	132	119	112
Conv(128,5,5)	5	100	5100	111	101	94	99
	ReLU 1	10	5100	135	133	142	143
Conv(64,5,5)	5	100	5100	130	102	105	108
	ReLU 1	10	5400	132	145	152	115
FC(30	**5**	**100**	**5400**	111	115	**69**	**71**
	LU 1	10	5400	145	141	121	119
FC(128)	5	100	5400	109	105	114	116

On Corrupted CIFAR-10 with $^1\epsilon = 0.05$; $^2\epsilon = 0.20$							
Models	τ	γ	U	IID^1	IID^2	1-C^1	5-C^2
	1	10	6500	89	95	110	128
Conv(	**5**	**100**	**6500**	**65**	**69**	110	113
	ReLU 1	10	6500	102	119	165	116
Conv(	**5**	**100**	**6500**	135	115	**82**	**73**
	ReLU 1	10	6500	189	157	112	119
Conv(64,5,5)	5	100	6500	112	111	190	181
	ReLU 1	10	6500	110	119	123	113
FC(300)	5	100	6500	182	113	109	134
	LU 1	10	6500	109	120	119	140
FC(128)	5	100	6500	109	102	111	131

On Corrupted LSUN-Bedroom with $^1\mathscr{E} = 0.05$; $^2\mathscr{E} = 0.20$							
Models	τ	γ	U	IID^1	IID^2	1-C^1	5-C^2
	1	10	6500	182	198	201	206
Conv(	**5**	**100**	**5000**	**84**	**89**	153	139
	ReLU 1	10	5000	113	120	170	168
Conv(128,5,5)	5	100	5000	131	155	202	201
	ReLU 1	10	5000	105	100	113	130
Conv(64,5,5)	5	100	5000	113	119	202	211
	ReLU 1	10	5000	120	101	119	142
FC(300)	5	100	5000	182	142	102	112
	LU 1	10	5000	98	103	112	149
FC(12	**5**	**100**	**4800**	108	113	**49**	**52**
+ ReLU							

On Corrupted GTSRB with $^1\mathscr{E} = 0.05$; $^2\mathscr{E} = 0.20$							
Models	τ	γ	U	IID^1	IID^2	1-C^1	5-C^2
	1	10	6000	114	129	98	109
Conv(	**5**	**100**	**6000**	**56**	**60**	**69**	**60**
	ReLU 1	10	6500	128	148	107	110
Conv(128,5,5)	5	100	6500	111	109	97	109
	ReLU 1	10	6500	101	162	165	189
Conv(64,5,5)	5	100	6500	97	101	110	119
	ReLU 1	10	6500	119	163	178	102
FC(300)	5	100	6500	129	119	156	152
	LU 1	10	6500	129	112	109	134
FC(128)	5	100	6500	119	130	156	104
+ ReLU							